# Generating Functions 3

Siddharth K

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## Problem 1

Applying the technique from the textbook:

$$\Omega = \{0, 1, 2\},$$

$$M \cong \mathcal{Z} \times \mathrm{SEQ}_{\Omega}(M),$$

$$M(z) = z(1 + M(z) + M(z)^2),$$

$$z = \frac{M(z)}{1 + M(z) + M(z)^2},$$

$$m_n = \frac{1}{n}[u^{n-1}](1 + u + u^2)^n.$$

The contributions to the exponent come from $u$ and $u^2$ terms, so we choose $k$ $u^2$ terms and $n - 2k - 1$ $u$ terms so that the sum is $n$:

$$m_n = \frac{1}{n} \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \binom{n}{k} \binom{n-k}{n-2k-1}.$$

## Problem 2

Let $\mathcal{B}$ be the combinatorial class of binary trees and $\mathcal{C}$ be the combinatorial class of one vertex along with a sequence of unary vertices. So

$$\mathcal{C} = \mathcal{Z} \times \mathrm{SEQ}(\mathcal{Z}),$$

$$C(z) = \frac{z}{1 - z}.$$

Each vertex of the full binary tree can be replaced by one of these vertex + sequences, so

$$M \cong B \circ C,$$

$$M = B\left(\frac{z}{1 - z}\right).$$

Since

$$B(z) = \frac{1 - \sqrt{1 - 4z^2}}{2z},$$

we get

$$M(z) = \frac{1 - z - \sqrt{1 - 2z - 3z^2}}{2z}.$$

Unsumming should give the same answer as problem 1.

# Problem 12

To show that $B(z) = A'(z)$, consider the number of objects of $\mathcal{B}$ with size $n$. It must have come from an object of $\mathcal{A}$ with size $n+1$, since one of the atoms got replaced with $\varepsilon$, and there were $n+1$ different atoms to point to, so $b_n = (n+1)a_{n+1}$. Multiplying by $z^n$ and summing over $n$ gives

$$\sum_{n \geq 0} b_n z^n = \sum_{n \geq 0} (n+1) a_{n+1} z^n = \sum_{n \geq 0} \frac{d}{dz} a_{n+1} z^{n+1},$$

$$B(z) = A'(z).$$

To show the product rule, we form a bijection between $\mathbf{D}(\mathcal{AB})$ and $\mathcal{A} \times \mathbf{D}\mathcal{B} + \mathbf{D}\mathcal{A} \times \mathcal{B}$. Take an object from $\mathbf{D}(\mathcal{AB})$, which is an erase-pointed form of $(a, b)$ where $a \in \mathcal{A}$ and $b \in \mathcal{B}$ with sizes $n_1$ and $n_2$ respectively. Splitting into atoms, the object is of either of the two forms:

$$(a_1, \ldots, \varepsilon, \ldots, a_{n_1}, b_1, \ldots, b_{n_2}),$$

$$(a_1, \ldots, a_{n_1}, b_1, \ldots, \varepsilon, \ldots, b_{n_2}).$$

If the erase-pointed atom was part of $a$, then we can split the object into an erase pointed object of $a$ and the original $b$, giving an object from $\mathbf{D}\mathcal{A} \times \mathcal{B}$. Otherwise, by similar reasoning, we get an object from $\mathcal{A} \times \mathbf{D}\mathcal{B}$.

This works in reverse too: if we have either

$$(a_1, \ldots, \varepsilon, \ldots, a_{n_1}), \quad (b_1, \ldots, b_{n_2}),$$

or

$$(a_1, \ldots, a_{n_1}), \quad (b_1, \ldots, \varepsilon, \ldots, b_{n_2}),$$

we can combine them into one object $(a_1, \ldots, b_{n_2})$ with one of the atoms replaced by $\varepsilon$, which gives an object from $\mathbf{D}(\mathcal{AB})$. Therefore,

$$\mathbf{D}(\mathcal{AB}) \cong \mathcal{A} \times \mathbf{D}\mathcal{B} + \mathbf{D}\mathcal{A} \times \mathcal{B},$$

$$(AB)'(z) = A(z)B'(z) + A'(z)B(z).$$

# Problem 13

We form a bijection between $\mathbf{D}(\mathcal{A} \circ \mathcal{B})$ and $(\mathbf{D}\mathcal{A} \circ \mathcal{B}) \times \mathbf{D}\mathcal{B}$. On the left, we pair each atom of an object from $\mathcal{A}$ with an object from $\mathcal{B}$, and erase one of the atoms of the object from $\mathcal{B}$, so it comes in the form

$$((a_1, \beta_1), \ldots, (a_i, (b_1, \ldots, \varepsilon, \ldots, b_n)), \ldots, (a_k, \beta_k)).$$

To form the right side, we pair each atom of an erase-pointed object from $\mathcal{A}$ with an object from $\mathcal{B}$, and take an erase-pointed object from $\mathcal{B}$. So it comes in the form

$$((a_1, \beta_1), \ldots, \varepsilon, \ldots, (a_k, \beta_k)), \quad (b_1, \ldots, \varepsilon, \ldots, b_n).$$

To go from the LHS to the RHS, we replace $(b_1, \ldots, \varepsilon, \ldots, b_n)$ with $\varepsilon$ and save it separately. To go from the RHS to the LHS, we replace the $\varepsilon$ in the $(\mathbf{D}\mathcal{A} \circ \mathcal{B})$ object with the $\mathbf{D}\mathcal{B}$ object, so

$$\mathbf{D}(\mathcal{A} \circ \mathcal{B}) \cong (\mathbf{D}\mathcal{A} \circ \mathcal{B}) \times \mathbf{D}\mathcal{B},$$
$$(A(B(z))' = A'(B(z))B'(z).$$

# Problem 14

Define

$$Q(z) = P(z) - 1 = \sum_{n \geq 1} n! z^n$$

where $P(z)$ is the generating function for permutations, so that

$$P(z) = 1 + Q(z),$$
$$I(z) = 1 - \frac{1}{P(z)} = 1 - \frac{1}{1 + Q(z)}$$
$$= 1 - \sum_{k \geq 0} (-1)^k Q(z)^k.$$

Then we extract the coefficient of $z^n$:

$$i_n = \sum_{k=1}^{n} (-1)^{k-1} [z^n] Q(z)^k,$$

$$[z^n] Q(z)^k = [z^n] \left( \sum_{j \geq 1} j! z^j \right)^k = \sum_{\substack{\sum_{i=1}^k n_i = n \\ n_i \geq 1}} \prod_{i=1}^{k} n_i!,$$

$$i_n = \sum_{k=1}^{n} (-1)^{k-1} \sum_{\substack{\sum_{i=1}^k n_i = n \\ n_i \geq 1}} \prod_{i=1}^{k} n_i!$$

$$= n! - \sum_{\substack{n_1 + n_2 = n \\ n_1, n_2 \geq 1}} n_1! n_2! + \sum_{\substack{n_1 + n_2 + n_3 = n \\ n_1, n_2, n_3 \geq 1}} n_1! n_2! n_3! - \cdots.$$

## Problem 15

Define $\mathcal{R}$ as the combinatorial class of lattice paths in $\mathbb{Z}^2$ that return to the origin at the end, but also possibly anytime in between. We have that

$$R(z) = \sum_{m\geq 0}\sum_{n\geq 0}\binom{2m+2n}{2m}\binom{2m}{m}\binom{2n}{n},$$

since out of all horizontal movements, we need to choose half to be right and half to be left, same for vertical movements, and out of $2m+2n$ total movements we need to choose $2m$ of them to be horizontal. Then, we have that

$$\mathcal{R} = \text{SEQ}(\mathcal{L}),$$

since any lattice path that ends at the origin but possibly crosses it in between can be broken into lattice paths that only cross the origin at the end. Isolating $L(z)$ gives

$$L(z) = 1 - \left(\sum_{m\geq 0}\sum_{n\geq 0}\binom{2m+2n}{2m}\binom{2m}{m}\binom{2n}{n}\right)^{-1}.$$